

Xudong Liao

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Education

- Sept. 2020 - **Hong Kong University of Science and Technology**, *Doctor of Philosophy*, Computer Science and Engineering
Oct. 2025 (expected) ○ Advisor: Prof. Kai Chen
- Sept. 2016 - **Wuhan University**, *Bachelor in Engineering*, Software Engineering
Jun. 2020 ○ Advisor: Prof. Yanjiao Chen
○ **Outstanding Graduate**

Research Interests

- Machine learning systems
- Congestion control
- Optical networks
- Datacenter networking

Selected Publications

* denotes equal contribution.

For the full list of publications, please refer to my Google Scholar page.

- [1] **Xudong Liao**, Yijun Sun, Han Tian, Xinchun Wan, Yilun Jin, Zilong Wang, Zhenghang Ren, Xinyang Huang, Wenxue Li, Kin Fai Tse, Zhizhen Zhong, Guyue Liu, Ying Zhang, Xiaofeng Ye, Yiming Zhang, and Kai Chen. MixNet: A Runtime Reconfigurable Optical-Electrical Fabric for Distributed Mixture-of-Experts Training. In *ACM SIGCOMM*, 2025.
- [2] **Xudong Liao**, Han Tian, Xinchun Wan, Chaoliang Zeng, Hao Wang, Junxue Zhang, Mengyu Ma, Guyue Liu, and Kai Chen. Towards Optimal Rack-scale μ s-level CPU Scheduling through In-Network Workload Shaping. In *USENIX ATC*, 2025.
- [3] Zhenghang Ren, Yuxuan Li, Zilong Wang, Xinyang Huang, Wenxue Li, Kaiqiang Xu, **Xudong Liao**, Yijun Sun, Bowen Liu, Han Tian, Junxue Zhang, Mingfei Wang, Zhizhen Zhong, Guyue Liu, Ying Zhang, and Kai Chen. Enabling Efficient GPU Communication over Multiple NICs with FuseLink. In *USENIX OSDI*, 2025.
- [4] Han Tian, **Xudong Liao**, Decang Sun, Chaoliang Zeng, Yilun Jin, Junxue Zhang, Xinchun Wan, Zilong Wang, Yong Wang, and Kai Chen. Achieving Fairness Generalizability for Learning-based Congestion Control with Jury. In *ACM EuroSys*, 2025.
- [5] **Xudong Liao***, Han Tian*, Chaoliang Zeng, Xinchun Wan, and Kai Chen. Astraea: Towards Fair and Efficient Learning-based Congestion Control. In *ACM EuroSys*, 2024.
- [6] Chaoliang Zeng*, **Xudong Liao***, Xiaodian Cheng, Han Tian, Xinchun Wan, Hao Wang, and Kai Chen. Accelerating Neural Recommendation Training with Embedding Scheduling. In *USENIX NSDI*, 2024.
- [7] Yiqing Ma, Han Tian, **Xudong Liao**, Junxue Zhang, Weiyan Wang, Kai Chen, and Xin Jin. Multi-Objective Congestion Control. In *ACM EuroSys*, 2022.
- [8] Han Tian*, **Xudong Liao***, Chaoliang Zeng, Junxue Zhang, and Kai Chen. Spine: An Efficient DRL-Based Congestion Control with Ultra-Low Overhead. In *ACM CoNEXT*, 2022.

Research Projects

2025 **MixNet: A Runtime Reconfigurable Optical-Electrical Fabric for MoE Training**, *SIGCOMM'25*

MixNet pioneers the first GPU interconnect that reconfigures its topology on-the-fly to match the highly localized, iteration-to-iteration dynamic traffic of Mixture-of-Experts (MoE) models. Its key innovation is a regionally reconfigurable OCS layer: because every expert-parallel (EP) group in MoE communicates mostly within a small, stable subset of GPUs, MixNet confines fast ($< \text{ms}$) optical circuit switching to those regions instead of the entire cluster. As a result, MixNet augments conventional electrical networks with a regionally reconfigurable OCS domain that (i) tracks per-iteration all-to-all demands, (ii) applies a greedy topology algorithm to provision just the optical circuits needed, and (iii) is driven by a customized NCCL runtime that hides sub-millisecond switch latency behind GPU compute. The regional OCS design removes the scalability bottleneck of traditional global OCS fabrics while retaining efficient performance. A prototype with 32 A100 GPUs and a Polaris OCS demonstrates the feasibility. Large-scale simulations with up to 30K GPUs show that MixNet improves “performance-per-dollar” by $1.2\text{--}1.5 \times$ (100 Gbps) and $1.9\text{--}2.3 \times$ (400 Gbps), enabling a scalable, cost-efficient path for future large-scale MoE deployments.

2024 **Herald: A Real-time Embedding Scheduler for Distributed DLRM Training**, *NSDI'24*

Herald is a real-time embedding scheduler that turns predictable, infrequently accessed embeddings into a communication-aware scheduling problem. Herald’s Location-Aware Input Allocation (LAIA) assigns each mini-batch to the GPU most likely to hit its cached embeddings, while a communication planner synchronizes only the embeddings required by other workers in the next iteration. Implemented on top of HET and evaluated on real-world DLRM workloads, Herald cuts embedding traffic by 48% - 89% and boosts end-to-end training throughput by up to $2.11 \times$ (TCP) / $1.61 \times$ (RDMA) over a 128-GPU cluster with 100 Gbps networks—all without sacrificing model accuracy.

2022-2025 **ML-Optimized Internet Congestion Control: From MOCC to Jury**, *EuroSys'22, CoNEXT'22, EuroSys'24, EuroSys'25*

This line of work explores RL-based congestion control protocols with a focus on practicality, fairness, and efficiency. MOCC introduces multi-objective RL to balance throughput, delay, and loss. Spine addresses the high inference overhead of RL-based CC by adopting a hierarchical design: a lightweight kernel-level executor reacts to ACKs, while a DRL agent periodically updates sub-policies. Astraea further advances this line by using centralized multi-agent reinforcement learning to directly optimize for convergence properties such as fairness, stability, and speed, achieving near-optimal Jain fairness (0.991) and $8.4\times$ faster convergence than prior work. Jury generalizes these insights to heterogeneous environments with theoretical guarantees. Together, this body of work demonstrates how learning-based CC can achieve robust, low-overhead, and fair network performance across a wide range of conditions.

Awards

2024	Travel Grant for NSDI'24 and EuroSys'24	<i>USENIX Association; ACM</i>
2020-2024	Postgraduate Studentship	<i>HKUST</i>
2020	Outstanding Graduate	<i>Wuhan University</i>
2017-2019	Second-class Scholarship	<i>Wuhan University</i>

Professional Experience

2025	Reviewer of IEEE/ACM Transactions on Networking
2023	Web chair of 7th Asia-Pacific Workshop on Networking (APNet'23)
2022-2025	Review Helper for CoNEXT'22, APNet'22, INFOCOM'22, NSDI'25 and SIGCOMM'25

Skills

Languages	Mandarin Chinese (native), English
Programming	C/C++, Python, $\text{\LaTeX}$, Bash scripts, RDMA verbs